

Bootstrap

❑ **Bootstrap 5** is a free and open-source CSS framework that is used for building responsive and mobile-first websites. It was developed and maintained by **Twitter**.

❑ Bootstrap 5 was released in 2021 and first availability of Bootstrap is 2011

❑ Framework is a collection of ready-made "classes" (selectors), which can be used in WebPages to apply styles to the elements easily.

❑ Bootstrap can be used without CSS knowledge; to customize the BS styles CSS & JS knowledge is required.

❑ Bootstrap was developed based on "Javascript" and "Popper"

Why Bootstrap?

1. It is a free front-end framework for faster and easier web development.
2. It combines HTML, CSS and JS based design templates for typography, forms, buttons, tables, navigation, image carousels and numerous other components, as well as JavaScript plug-in.
3. It provides you with a convenient approach for creating responsive designs.
4. It is supported by all popular browsers.

What is Responsive Web Design?

Responsive web design is about creating web sites which automatically adjust themselves to look good on all devices, from small phones to large desktops.

Bootstrap Environment preparation (with downloads)

Download BS:

- > go to <https://getbootstrap.com> (select bootstrap5, it downloads .zip file)
- > extract that zip into E:\BOOTSTRAP folder

Download Popper:

- > go to <https://unpkg.com/popper.js> (select Download the uncompressed, development jQuery)
- > it open source code then save that with a name "popper.js" in E:\BOOTSTRAP folder

Importing (linker):

```
<link rel="stylesheet" href="bootstrap.css">
<script src="popper.js"></script>
<script src="bootstrap.js"></script>
```

Bootstrap 5 CDN (without downloads)

If you don't want to download and host Bootstrap 5 yourself, you can include it from a CDN (Content Delivery Network).

jsDelivr provides CDN support for Bootstrap's CSS and JavaScript:

CSS only

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet"
integrity="sha384-QWTKZyjpPEjISv5WaRU9OFeRpok6YctnYmDr5pNlyT2bRjXh0JMhY6hW+ALEwIH"
crossorigin="anonymous">
```

JS, Popper.js, and jQuery

```
<script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"
integrity="sha384-DfXdz2htPH0lsSSs5nCTpuj/zy4C+OGpamoFVy38MVBnE+IbbVYUew+OrCXaRkfj"
crossorigin="anonymous"></script>
```

```
<script src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.11.8/dist/umd/popper.min.js"
integrity="sha384-l7E8VVD/ismYTF4hNIPjVp/Zjvgyol6VFvRkX/vR+Vc4jQkC+hVqc2pM8ODewa9r"
crossorigin="anonymous"></script>
```

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.min.js"
integrity="sha384-0pUGZvbkm6XF6gxjEnlmuGrJXVbNuzT9qBBavbLwCsOGabYfZoOT0to5e9ruptLy"
crossorigin="anonymous"></script>
```

Warning: All the files listed above have to be linked in the given order.

<https://getbootstrap.com/>

For example: `css/mdb.min.css` must be always linked after `css/bootstrap.min.css`, and `js/jquery-3.3.1.min.js` must be always linked before `js/bootstrap.min.js`, etc.

Note:

The attributes `integrity` and `crossorigin` have been added to CDN links to implement **Subresource Integrity** (SRI). It is a security feature that enables you to mitigate the risk of attacks originating from compromised CDNs, by ensuring that the files your website fetches (from a CDN or anywhere) have been delivered without unexpected or malicious modifications. It works by allowing you to provide a cryptographic hash that a fetched file must match.

HTML5 doctype

Bootstrap requires the use of the HTML5 doctype.

Without it, you'll see some funky incomplete styling, but including it shouldn't cause any considerable hiccups.

Viewport meta

Bootstrap is developed *mobile first*, a strategy in which we optimize code for mobile devices first and then scale up components as necessary using media queries. To ensure proper rendering and touch zooming for all devices, add the responsive viewport meta tag to your `<head>`.

Bootstrap Containers

Containers are the most basic layout element in Bootstrap and are required when using the grid system. Containers are basically used to wrap content with some padding. They are also used to align the content horizontally center on the page in case of fixed width layout.

Bootstrap provides three different types of containers:

- `.container`, which has a max-width at each responsive breakpoint.
- `.container-fluid`, which has 100% width at all breakpoints.
- `.container-{breakpoint}`, which has 100% width until the specified breakpoint.